FINAL PROJECT REPORT

Secure Bank Branch Network

Design and Implementation of a Secure Multi-Branch Bank Network in Cisco Packet Tracer

Abstract

This project presents the design, configuration and testing of a secure multi-branch banking network built in Cisco Packet Tracer. The network connects one Head Office to five branch offices and shows how core enterprise networking technologies work together to provide reliable connectivity, centralized services and controlled access.

The Head Office is divided into Admin, Employee, IT and Server VLANs, while each branch has Staff, ATM and Guest VLANs. Inter-VLAN communication is provided by Router-on-a-Stick, using one physical router link with a subinterface as the default gateway of each VLAN. OSPF (Process ID 1, Area 0) dynamically exchanges routes between the Head Office and all branches. DHCP assigns IP settings automatically, and a central Bank Server (192.168.40.10) at the Head Office provides DNS, resolving bank.local, and HTTP services.

Security is enforced in layers: VLANs separate users and departments, extended ACLs implement the bank's access policy between networks, Port Security with sticky MAC addresses protects switch access ports, and SSH provides encrypted remote router management. The network was verified using ping, ipconfig, show commands and browser tests. The project is a simulation and educational model, not a production banking network.

Keywords: VLAN, Router-on-a-Stick, OSPF, DHCP, DNS, ACL, Port Security, SSH, Cisco Packet Tracer

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1. Introduction

Banks depend on networks that are secure, reliable and centrally manageable. A bank typically operates a Head Office and several branches, and each branch must reach central services while staff, ATM machines and customers (guests) remain properly separated from one another.

This project builds such a network in Cisco Packet Tracer. One Head Office is connected to five branches (Branch 01 to Branch 05), giving six sites in total. Eight networking technologies are combined: VLAN, Inter-VLAN routing, DHCP, DNS, OSPF, ACL, SSH and Port Security.

2. Problem Statement

A multi-branch bank network faces three main problems:

- Branches must be able to reach central services at the Head Office.
- Unauthorized traffic between user groups must be blocked.
- A flat network lacks segmentation and security, so any device could reach any other device.

3. Aim and Objectives

Aim: To build a secure multi-branch bank network that is segmented with VLANs, routed with OSPF, and protected with ACLs, Port Security and SSH.

Objectives:

1. Design a Head Office plus five-branch topology.
2. Separate departments with VLANs.
3. Enable inter-VLAN routing using Router-on-a-Stick.
4. Run OSPF Area 0 across all sites.
5. Assign IP settings automatically with DHCP.
6. Host DNS and HTTP on a central Bank Server.
7. Control access between networks with ACLs.
8. Manage routers securely with SSH.
9. Protect access ports with Port Security.
10. Verify the network with ping, show commands and a browser.

4. Project Scope

Area	Description
Head Office	Admin, Employee, IT and Server VLANs
Branches	Branch 01 to Branch 05
Branch VLANs	Staff, ATM and Guest networks
Addressing	One subnet and gateway per VLAN
Routing	OSPF Process 1, Area 0
Switching	Access ports, trunk ports, Port Security
Services	DHCP, DNS and HTTP on the Bank Server
Security	ACL, SSH and Port Security
Testing	ping, show commands and browser test

The network devices used are routers, switches, PCs, ATM endpoints and one central bank server.

5. Network Topology and Components

The Head Office router sits at the centre of the design. It connects to the Head Office VLANs, to the Bank Server, and to each of the five branch routers. All six sites participate in OSPF Area 0.

Component	Count	Details
Sites	6	Head Office and 5 branches
Head Office VLANs	4	Admin, Employee, IT, Server
Branch VLANs	3 per branch	Staff, ATM, Guest
Bank Server	1	DNS and HTTP at 192.168.40.10

Protocols and features used: IPv4, VLAN, Inter-VLAN routing, OSPF, DHCP, DNS, HTTP, ACL, SSH and Port Security.

6. IP Addressing and VLAN Plan

Every VLAN has its own subnet and its own default gateway. This lets routers forward packets to the correct destination and lets ACLs control traffic between groups.

Location	VLAN	Purpose	Network
Head Office	10	Admin	192.168.10.0/24
Head Office	20	Employee	192.168.20.0/24
Head Office	30	IT	192.168.30.0/24
Head Office	40	Server	192.168.40.0/24
Branch 01	10	Staff	192.168.50.0/24
Branch 01	20	ATM	192.168.60.0/24
Branch 01	30	Guest	192.168.70.0/24
Branches 02 to 05	10	Staff	Branch-specific subnet
Branches 02 to 05	20	ATM	Branch-specific subnet
Branches 02 to 05	30	Guest	Branch-specific subnet

VLANs separate departments, reduce unnecessary broadcast traffic and support different security policies for different user groups.

7. Switching and Inter-VLAN Routing

7.1 Access and Trunk Ports

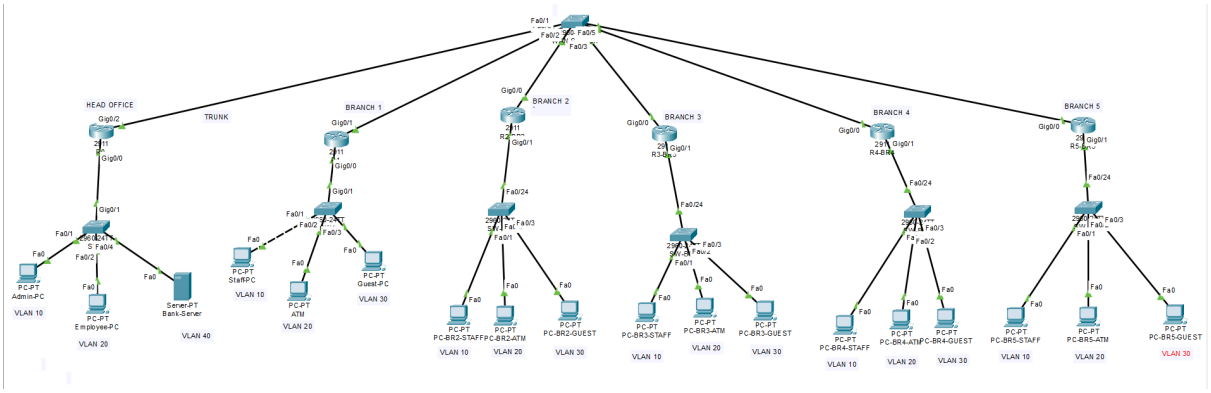
An access port connects an endpoint such as a PC or ATM to a single VLAN. A trunk port carries the traffic of several VLANs over one physical link using VLAN tags. Access ports connect endpoints; trunk ports connect switches to routers and to other switches.

7.2 Router-on-a-Stick

Router-on-a-Stick uses a single physical router interface divided into subinterfaces. Each subinterface belongs to one VLAN and serves as that VLAN's default gateway. The branch switch is connected to the branch router by one trunk link. For Branch 01 the gateways are:

VLAN	Network	Gateway
Staff	192.168.50.0/24	192.168.50.1
ATM	192.168.60.0/24	192.168.60.1
Guest	192.168.70.0/24	192.168.70.1

This enables inter-VLAN routing while using only one physical router link.



8. Dynamic Routing with OSPF

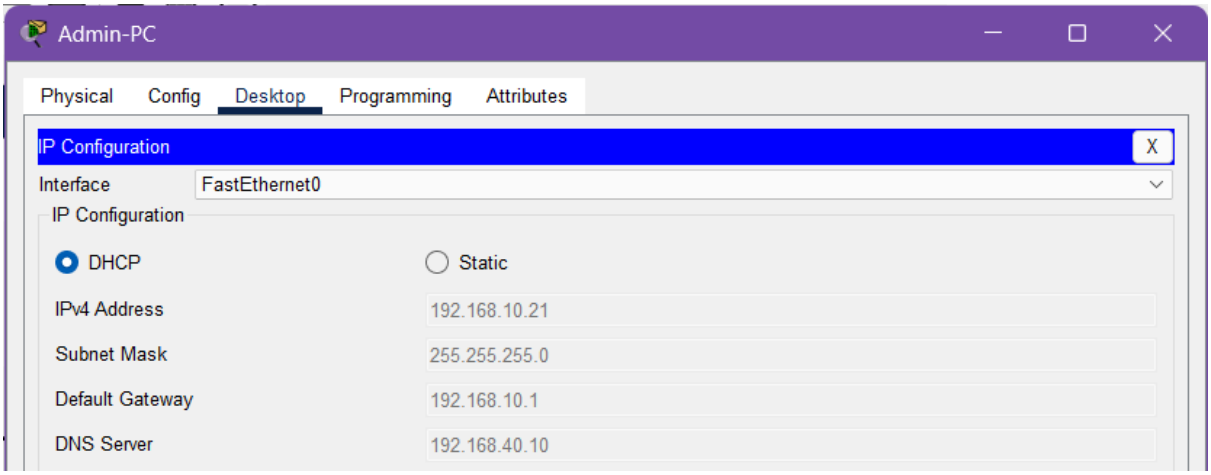
OSPF (Open Shortest Path First) is a dynamic routing protocol in which routers exchange routing information and learn remote networks automatically. In this project OSPF Process ID 1 and Area 0 are used across the Head Office and all five branches.

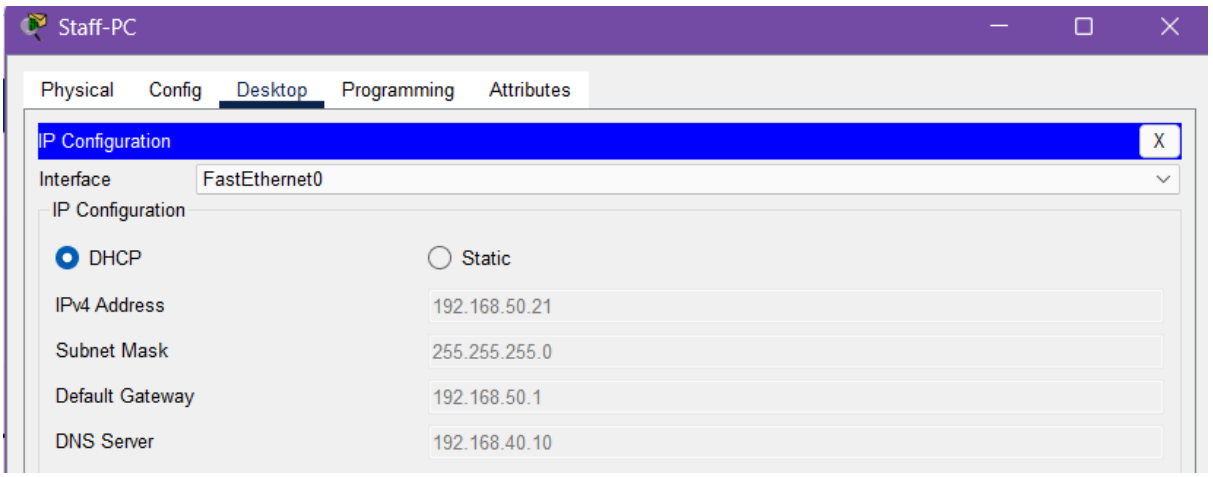
OSPF reduces the need for manual static routes and shares routing updates automatically. Routers first form neighbor relationships and only then exchange routing information, which is why the neighbor state is checked during verification.

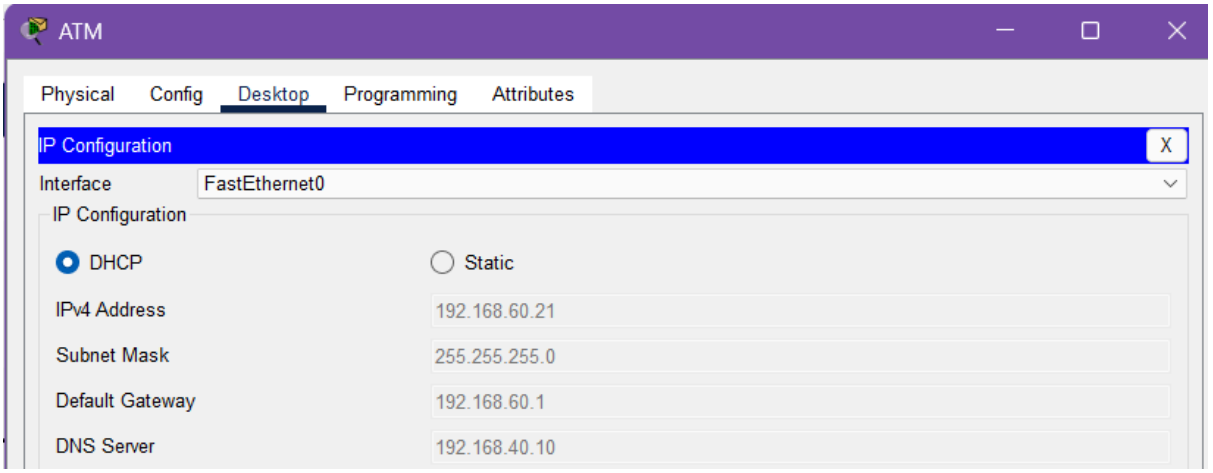
9. Network Services: DHCP, DNS and HTTP

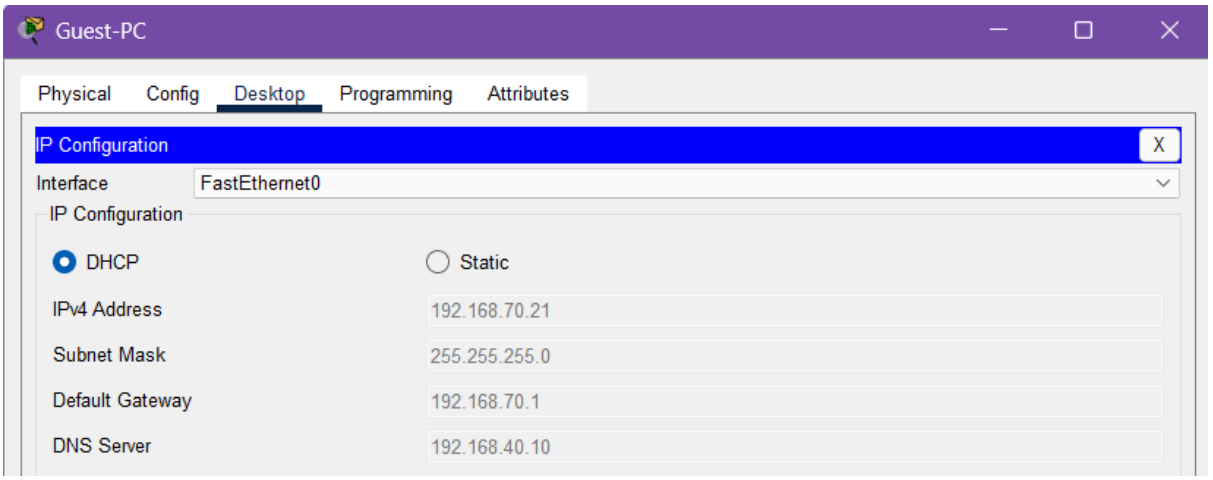
9.1 DHCP

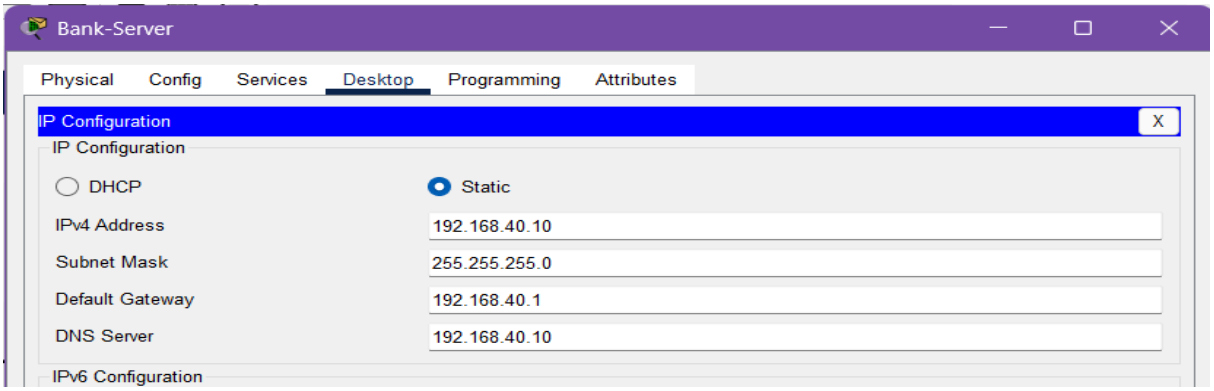
DHCP automatically assigns an IP address, subnet mask, default gateway and DNS server to endpoints. Pools were configured for the Staff, ATM and Guest networks. The process follows four steps: Discover, Offer, Request and Acknowledge (DORA). DHCP reduces manual configuration and helps prevent addressing mistakes




Staff-PC
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.50.21
Subnet Mask
255.255.255.0
Default Gateway
192.168.50.1
DNS Server
192.168.40.10

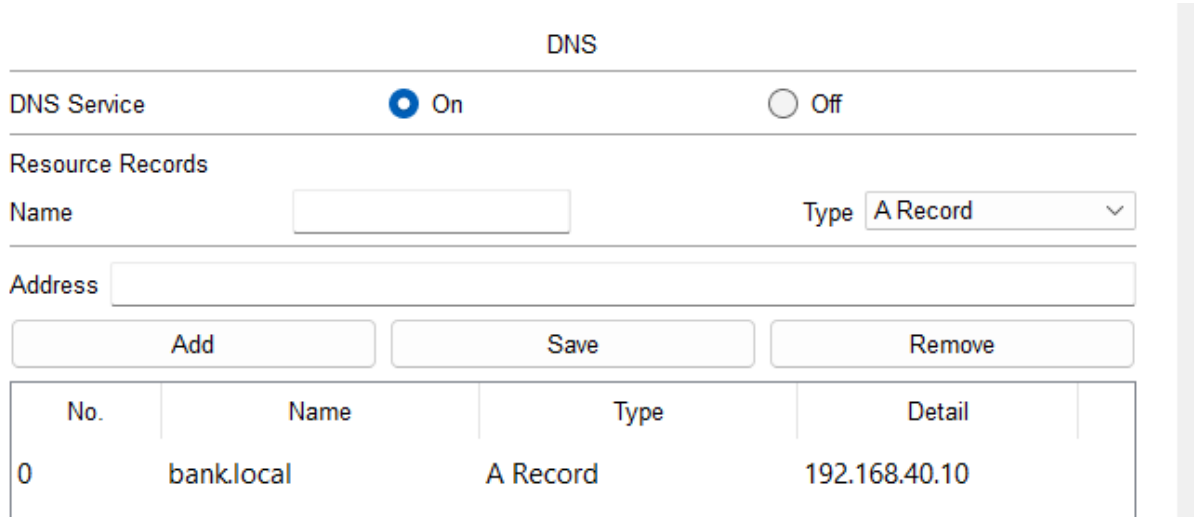

ATM
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.60.21
Subnet Mask
255.255.255.0
Default Gateway
192.168.60.1
DNS Server
192.168.40.10


Guest-PC
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.70.21
Subnet Mask
255.255.255.0
Default Gateway
192.168.70.1
DNS Server
192.168.40.10


Bank-Server
Physical
Config
Services
Desktop
Programming
Attributes
IP Configuration
X
IP Configuration
DHCP
Static
IPv4 Address
192.168.40.10
Subnet Mask
255.255.255.0
Default Gateway
192.168.40.1
DNS Server
192.168.40.10
IPv6 Configuration

9.2 DNS

DNS translates names into IP addresses. The DNS service on the central server resolves **bank.local** to **192.168.40.10**, so users can type a name instead of remembering an IP address. DNS only provides name resolution; it does not authorize access, which remains the job of the ACLs.



9.3 HTTP and the Central Bank Server

The Bank Server is located at the Head Office (192.168.40.10) and provides DNS and HTTP. Authorized users can open the bank web page in a browser, and ACLs decide which networks may reach the server.



10. Security Implementation

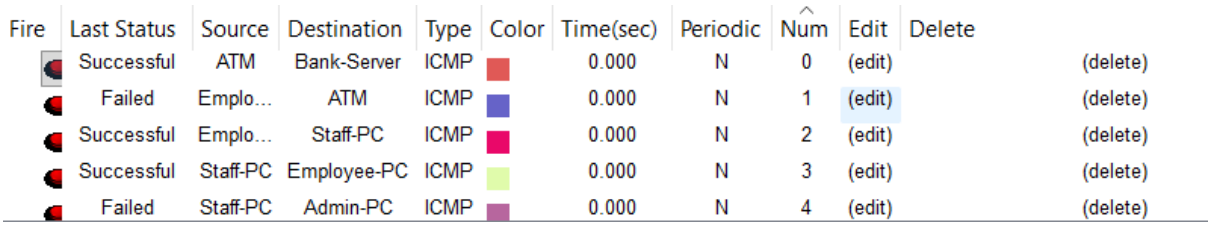
Security is applied in four layers:

- VLANs separate departments and users.
- ACLs filter traffic between networks.
- Port Security protects switch access ports.
- SSH secures router administration.

10.1 Access Control Lists

Extended ACLs filter traffic by source IP, destination IP and protocol. ACLs are read from top to bottom and the first matching rule applies. Traffic that matches no permit rule is blocked by the implicit deny at the end.

Source	Destination	Action
HO Admin	Branches, Bank Server, HO Employee	Permit
HO Employee	Branch Staff	Permit
HO Employee	Branch ATM and Guest	Deny
Branch Staff	HO Employee, own-branch Guest, Bank Server	Permit
Branch ATM	Own-branch Guest, Bank Server	Permit
Branch Guest	Bank Server and other branches	Deny
Branch Staff	HO Admin	Deny
Branch ATM	HO Admin	Deny

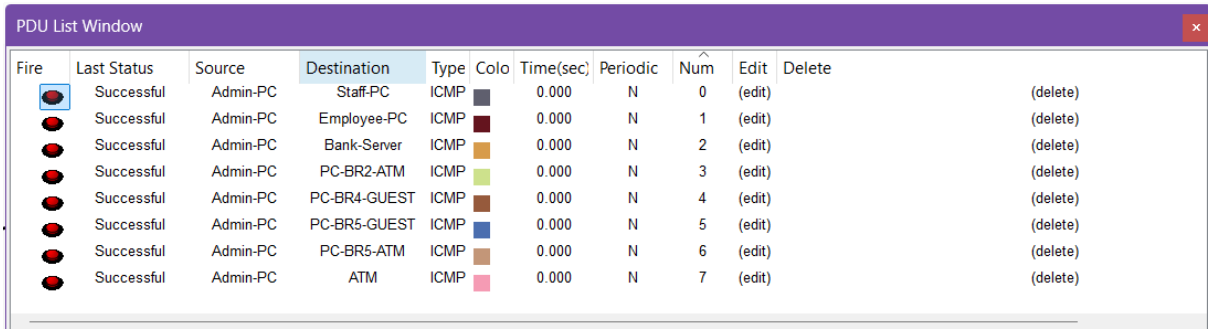


Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	ATM	Bank-Server	ICMP		0.000	N	0	(edit)	(delete)
	Failed	Emplo...	ATM	ICMP		0.000	N	1	(edit)	(delete)
	Successful	Emplo...	Staff-PC	ICMP		0.000	N	2	(edit)	(delete)
	Successful	Staff-PC	Employee-PC	ICMP		0.000	N	3	(edit)	(delete)
	Failed	Staff-PC	Admin-PC	ICMP		0.000	N	4	(edit)	(delete)

Where configured, a narrow DNS-only exception may allow Guest DNS requests to the server while all other Guest access remains blocked.

10.2 Expected Connectivity Matrix

Source	Destination	Expected Access
HO Admin	All Branch Staff	Allow
HO Admin	All Branch ATM	Allow
HO Admin	All Branch Guest	Allow
HO Employee	Branch Staff	Allow
Branch Staff	HO Employee	Allow
HO Employee	Branch ATM	Deny
HO Employee	Branch Guest	Deny
Staff	Own Branch Guest	Allow
Own Branch Guest	Staff	Allow
ATM	Own Branch Guest	Allow
Own Branch Guest	ATM	Allow
Branch Staff	Bank Server	Allow
Branch ATM	Bank Server	Allow
Branch Staff	HO Admin	Deny
Branch ATM	HO Admin	Deny
Branch Guest	Other Branch	Deny

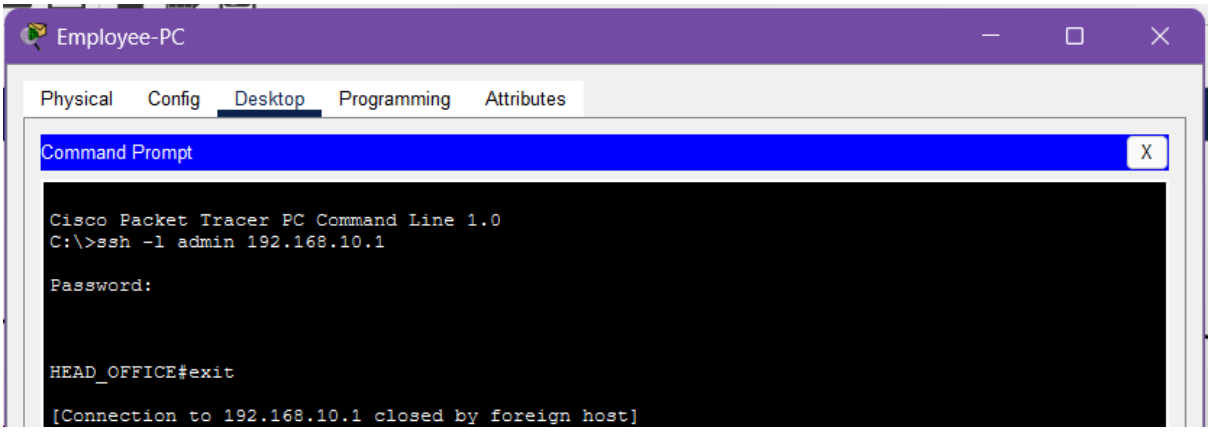


10.3 Port Security

Port Security limits which MAC addresses may use a switch access port. With sticky MAC, the switch learns a device's MAC address and saves it. It was configured on selected access ports to prevent unauthorized devices from connecting.

10.4 SSH

SSH provides encrypted remote management of the routers, so an administrator can configure devices remotely without visiting each one and without sending credentials in clear text.



11. Traffic Flow Example

The path of an authorized request from a Branch Staff PC to the Bank Server:

1. **DHCP** assigns the PC its IP address, gateway and DNS server.
2. **DNS** resolves bank.local to 192.168.40.10.
3. The PC sends the request to its **default gateway**, the branch router.
4. The branch router performs the **ACL** check and permits authorized traffic.
5. **OSPF** routes guide the packets toward the Head Office.
6. The Head Office router forwards the traffic to the Bank Server.
7. The server responds, and return traffic follows routing and ACL policy.

12. Testing and Verification

Command / Test	What It Checks
ping	IP reachability
ipconfig	PC IP configuration
show ip route	Router routing table
show ip ospf neighbor	OSPF neighbor state
show access-lists	ACL rules and match counters
show vlan brief	VLAN configuration
show port-security	Port Security state
Browser / DNS lookup	HTTP access and name resolution

```
C:\>ping 192.168.50.21

Pinging 192.168.50.21 with 32 bytes of data:

Reply from 192.168.50.21: bytes=32 time<1ms TTL=126
Reply from 192.168.50.21: bytes=32 time<1ms TTL=126
Reply from 192.168.50.21: bytes=32 time<1ms TTL=126
Reply from 192.168.50.21: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.50.21:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

```
C:\>ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..: bank.local
   Link-local IPv6 Address.....: FE80::209:7CFF:FE18:766C
   IPv6 Address.....: ::
   IPv4 Address.....: 192.168.10.21
   Subnet Mask.....: 255.255.255.0
   Default Gateway.....: ::
                                 192.168.10.1

Bluetooth Connection:

   Connection-specific DNS Suffix..: bank.local
   Link-local IPv6 Address.....: ::
   IPv6 Address.....: ::
   IPv4 Address.....: 0.0.0.0
   Subnet Mask.....: 0.0.0.0
   Default Gateway.....: ::
                                 0.0.0.0

C:\>
```

```
HEAD_OFFICE>enable
HEAD_OFFICE#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.1.0/29 is directly connected, GigabitEthernet0/2
L       10.0.1.1/32 is directly connected, GigabitEthernet0/2
     192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, GigabitEthernet0/0.10
L       192.168.10.1/32 is directly connected, GigabitEthernet0/0.10
     192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.20.0/24 is directly connected, GigabitEthernet0/0.20
L       192.168.20.1/32 is directly connected, GigabitEthernet0/0.20
     192.168.30.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.30.0/24 is directly connected, GigabitEthernet0/0.30
L       192.168.30.1/32 is directly connected, GigabitEthernet0/0.30
     192.168.40.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.40.0/24 is directly connected, GigabitEthernet0/0.40
L       192.168.40.1/32 is directly connected, GigabitEthernet0/0.40
O    192.168.50.0/24 [110/2] via 10.0.1.6, 00:29:10, GigabitEthernet0/2
O    192.168.60.0/24 [110/2] via 10.0.1.6, 00:29:10, GigabitEthernet0/2
O    192.168.70.0/24 [110/2] via 10.0.1.6, 00:29:10, GigabitEthernet0/2
O    192.168.80.0/24 [110/2] via 10.0.1.2, 00:29:10, GigabitEthernet0/2
O    192.168.90.0/24 [110/2] via 10.0.1.2, 00:29:10, GigabitEthernet0/2
O    192.168.100.0/24 [110/2] via 10.0.1.2, 00:29:10, GigabitEthernet0/2
O    192.168.110.0/24 [110/2] via 10.0.1.3, 00:29:10, GigabitEthernet0/2
O    192.168.120.0/24 [110/2] via 10.0.1.3, 00:29:10, GigabitEthernet0/2
O    192.168.130.0/24 [110/2] via 10.0.1.3, 00:29:10, GigabitEthernet0/2
O    192.168.140.0/24 [110/2] via 10.0.1.4, 00:29:10, GigabitEthernet0/2
O    192.168.150.0/24 [110/2] via 10.0.1.4, 00:29:10, GigabitEthernet0/2
O    192.168.160.0/24 [110/2] via 10.0.1.4, 00:29:10, GigabitEthernet0/2
O    192.168.170.0/24 [110/2] via 10.0.1.5, 00:29:10, GigabitEthernet0/2
O    192.168.180.0/24 [110/2] via 10.0.1.5, 00:29:10, GigabitEthernet0/2
O    192.168.190.0/24 [110/2] via 10.0.1.5, 00:29:10, GigabitEthernet0/2

HEAD_OFFICE#
```

```
HEAD_OFFICE#show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
4.4.4.4           1   FULL/DROTHER    00:00:32    10.0.1.3        GigabitEthernet0/2
5.5.5.5           1   FULL/DROTHER    00:00:32    10.0.1.4        GigabitEthernet0/2
6.6.6.6           1   FULL/DROTHER    00:00:32    10.0.1.5        GigabitEthernet0/2
192.168.70.1      1   FULL/DR         00:00:32    10.0.1.6        GigabitEthernet0/2
3.3.3.3           1   FULL/DROTHER    00:00:32    10.0.1.2        GigabitEthernet0/2
HEAD_OFFICE#
```

```
BRANCH_01>en
BRANCH_01#enable
BRANCH_01#show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
192.168.40.1      1   FULL/BDR        00:00:37    10.0.1.1        GigabitEthernet0/1
4.4.4.4           1   FULL/DROTHER    00:00:37    10.0.1.3        GigabitEthernet0/1
5.5.5.5           1   FULL/DROTHER    00:00:37    10.0.1.4        GigabitEthernet0/1
6.6.6.6           1   FULL/DROTHER    00:00:37    10.0.1.5        GigabitEthernet0/1
3.3.3.3           1   FULL/DROTHER    00:00:37    10.0.1.2        GigabitEthernet0/1
BRANCH_01#`
```

```
SW1#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Gig0/1      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Gig0/1      1-1005

Port        Vlans allowed and active in management domain
Gig0/1      1,10,20,30

Port        Vlans in spanning tree forwarding state and not pruned
Gig0/1      1,10,20,30

SW1#
```

```
R2-BR2>enable
R2-BR2#show access-lists
Extended IP access list BR2_GUEST_FILTER
    10 permit udp any eq bootpc any eq bootps (3 match(es))
    20 permit udp 192.168.100.0 0.0.0.255 host 192.168.40.10 eq domain
    30 permit ip 192.168.100.0 0.0.0.255 192.168.80.0 0.0.0.255
    40 permit tcp 192.168.100.0 0.0.0.255 host 192.168.40.10 eq domain
    50 permit ip 192.168.100.0 0.0.0.255 192.168.90.0 0.0.0.255
    60 permit tcp 192.168.100.0 0.0.0.255 host 192.168.40.10 eq www
    70 permit ip 192.168.100.0 0.0.0.255 192.168.10.0 0.0.0.255
    80 deny ip 192.168.100.0 0.0.0.255 any
Extended IP access list BR2_ATM_FILTER
    10 permit udp any eq bootpc any eq bootps (2 match(es))
    20 permit ip 192.168.90.0 0.0.0.255 192.168.100.0 0.0.0.255
    30 permit ip 192.168.90.0 0.0.0.255 192.168.10.0 0.0.0.255 (2 match(es))
    40 permit ip 192.168.100.0 0.0.0.255 192.168.90.0 0.0.0.255
    50 permit ip 192.168.90.0 0.0.0.255 host 192.168.40.10
    60 permit ip host 192.168.40.10 192.168.90.0 0.0.0.255
    70 deny ip 192.168.90.0 0.0.0.255 any
Extended IP access list BR2_STAFF_FILTER
    10 permit udp any eq bootpc any eq bootps (3 match(es))
    20 permit ip 192.168.80.0 0.0.0.255 192.168.100.0 0.0.0.255
    30 permit ip 192.168.80.0 0.0.0.255 192.168.10.0 0.0.0.255 (2 match(es))
    40 permit ip 192.168.80.0 0.0.0.255 192.168.20.0 0.0.0.255
    50 permit ip 192.168.20.0 0.0.0.255 192.168.80.0 0.0.0.255
    60 permit ip 192.168.80.0 0.0.0.255 host 192.168.40.10
    70 permit ip host 192.168.40.10 192.168.80.0 0.0.0.255
    80 deny ip 192.168.80.0 0.0.0.255 any

R2-BR2#
```

```
SW1>enable
SW1#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/4, Fa0/5, Fa0/6, Fa0/7
                                                Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                                Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                                Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                                Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                                Fa0/24, Gig0/2
10   STAFF                            active    Fa0/1
20   ATM                              active    Fa0/2
30   GUEST                            active    Fa0/3
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
SW1#-
```

```
SW1#enable
SW1#show port-security
Secure Port MaxSecureAddr CurrentAddr SecurityViolation Security Action
                (Count)       (Count)         (Count)
------------------------------------------------------------------------
        Fa0/1        1           1                0         Restrict
        Fa0/2        1           1                0         Restrict
        Fa0/3        1           1                0         Restrict
------------------------------------------------------------------------
SW1#
```

Configuration commands confirm that features are set up correctly, while ping and browser tests confirm real connectivity. ACL match counters also show that permit and deny rules are being hit as intended.

13. Project Outcomes

- Head Office and five branches built.
- VLANs with Router-on-a-Stick routing.
- DHCP addressing and DNS (bank.local).
- OSPF Area 0 routing across all sites.
- ACL access policy between networks.
- Port Security and SSH management.
- Central Bank Server providing DNS and HTTP.

14. Limitations and Future Work

This project is a simulation and educational model, not a production banking network. Only concepts within the course scope and Packet Tracer are implemented. Real banks additionally use:

- Redundant firewalls and IDS/IPS
- VPN connections between sites
- Centralized authentication
- Monitoring and logging
- Encryption and high availability
- Regulatory and compliance controls

These can be considered as future improvements to the design.

15. Conclusion

The Secure Bank Branch Network shows how enterprise networking technologies work together. VLANs separate departments, Router-on-a-Stick enables inter-VLAN routing, DHCP and DNS simplify addressing and name resolution, OSPF shares routes between sites, and ACLs, Port Security and SSH protect access and management. The result is reliable connectivity, centralized services, easier management and controlled access across the Head Office and five branches.